

BASIC ENCRYPTION AND DECRYPTION USING CAESAR CIPHER



Internship Project Report

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Project Title: Basic Encryption and Decryption Using Caesar Cipher

Organization: Decodelabs

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1. Introduction

Cryptography is an essential part of cybersecurity that helps protect sensitive information from unauthorized access. Encryption converts readable data into an unreadable format, while decryption converts it back into its original form.

This project demonstrates the implementation of the Caesar Cipher, one of the simplest encryption techniques, using Python programming.

2. Objective

- To understand the basics of cryptography.
- To implement encryption and decryption techniques.
- To learn how data can be secured using algorithms.
- To gain practical experience in Python programming.

3. Technology Used

- Python 3
- Caesar Cipher Algorithm
- Command Line Interface (CLI)

4. Project Description

The Caesar Cipher works by shifting each letter in the plaintext by a fixed number of positions in the alphabet.

For example, with a shift value of 3:

- A becomes D
- B becomes E
- C becomes F

The program accepts user input and a shift value, then encrypts the text. It can also decrypt the encrypted text back to the original message.

5. Algorithm

Encryption

1. Accept text from the user.
2. Accept a shift value.
3. Shift each alphabet character by the given value.
4. Display the encrypted text.

Decryption

1. Take the encrypted text.
2. Apply the reverse shift.
3. Display the original text.

6. Source Code

#Basic Encryption and Decryption

```
def encrypt(text, shift):
    result = ""
    for char in text:
        if char.isalpha():
            start = ord('A') if char.isupper() else ord('a')
            result += chr((ord(char) - start + shift) % 26 + start)
        else:
            result += char
    return result

def decrypt(text, shift):
    return encrypt(text, -shift)

# User Input
message = input("Enter text: ")
shift = int(input("Enter shift value: "))
```

```
encrypted = encrypt(message, shift)

decrypted = decrypt(encrypted, shift)

print("\nEncrypted Text:", encrypted)

print("Decrypted Text:", decrypted)
```

7. Sample Output

Input:

Hello World

Shift Value:

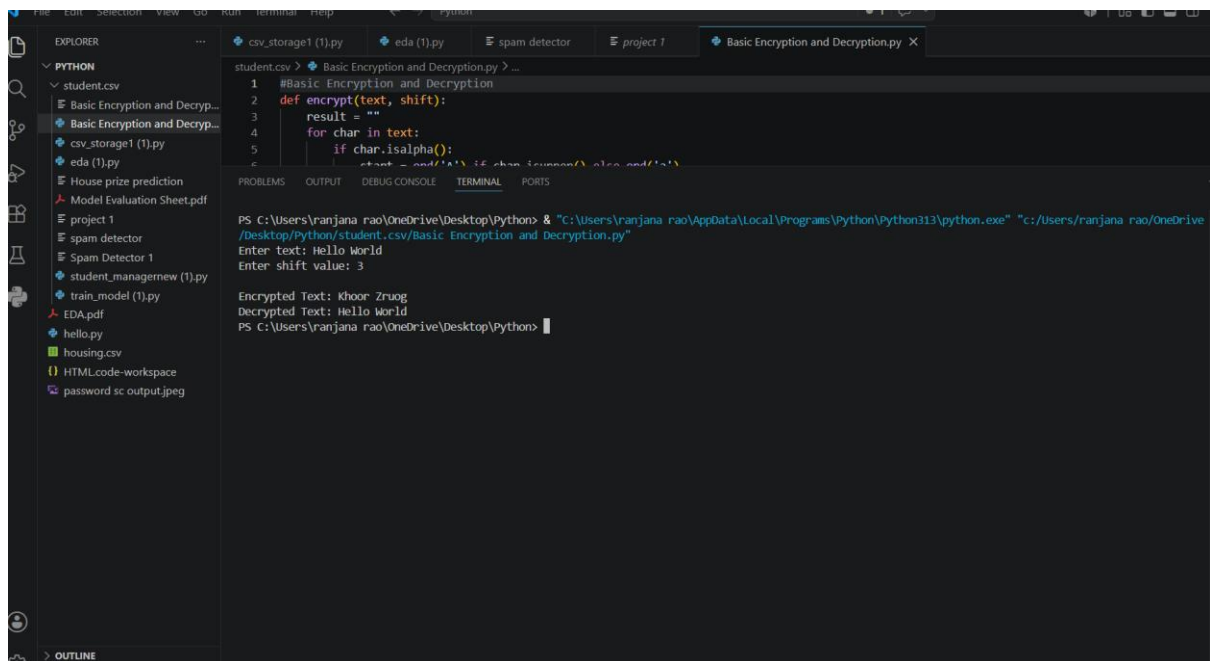
3

Output:

Encrypted Text: Kloor Zruog

Decrypted Text: Hello World

8. Sample Output Screenshot

A screenshot of a Python IDE interface. The Explorer pane on the left shows a project named 'student.csv' containing files like 'Basic Encryption and Decryption.py', 'csv_storage1 (1).py', and 'eda (1).py'. The main editor window displays the code for 'Basic Encryption and Decryption.py', which includes a function 'encrypt(text, shift)' and a function 'decrypt(text, shift)'. The terminal pane at the bottom shows the command prompt running the script, with input for 'Enter text: Hello World' and 'Enter shift value: 3', followed by the output 'Encrypted Text: Kloor Zruog' and 'Decrypted Text: Hello World'.

9. Advantages

- Simple to implement.
- Helps understand basic cryptography concepts.
- Useful for learning encryption techniques.
- Beginner-friendly cybersecurity project.

10. Limitations

- Not secure for real-world applications.
- Can be cracked easily using brute force methods.
- Suitable only for educational purposes.

11. Conclusion

The project successfully demonstrates the basic concepts of encryption and decryption using the Caesar Cipher algorithm. It provides a practical understanding of cryptography and serves as a foundation for learning advanced cybersecurity concepts and encryption techniques.

12. Learning Outcomes

- Understanding of Cryptography
- Encryption and Decryption Concepts
- Python Programming Skills
- Cybersecurity Fundamentals
- Problem-Solving and Logical Thinking

13. References

1. Python Official Documentation
2. Cryptography Fundamentals
3. Cybersecurity Learning Resources